

5. ARES: The Mars Port (Phase 4)

Mars atmosphere is 0.6% of Earth's. The hard parts of the Earth machine are nearly free there.

The tube holds back 0.6% of the pressure the Earth tube fights; the plasma window's job is 99.4% done by the planet; exit heating is a fraction of the Earth case. Dust storms, not aerodynamics, are the main environmental design driver.

Element	Figure	Notes
Orbit track	~3.4 km/s → ~180 km @ 3.3 g	105-second run to low Mars orbit
Escape track	~5.0 km/s → ~390 km @ 3.3 g	155-second run; between Moon and Earth in scale, far easier than Earth in engineering
Energy	1.6 / 3.5 kWh per kg	orbit / escape
Staging	Phobos & Deimos	Catch points and propellant depots — the warehouse district; SEP tugs fly the cruise

Every Mars mission arrives needing return propellant. ISRU solves it on the surface (Sabatier methalox from CO₂ and water ice); ARES solves it in orbit — propellant rides to Phobos at electromagnetic cost and waits in a depot for any departing vehicle. Return cargo, manufactured exports, and samples ride the same track.

ARES makes Mars a place you can leave as easily as you arrive.

The Moon as deep-space launch platform. A Mars-transfer from the lunar surface costs ~3.0 km/s (≈1.25 kWh/kg) — roughly one-fifteenth the energy of the same trip from Earth's surface. Anything Mars-bound that can be built on, or staged through, the Moon ships from the lunar track: relay satellites, sensor networks, pre-positioned supplies. Starship carries crew and cargo that must arrive fresh; the mass drivers send everything else ahead, early and cheap. The Mars supply chain runs through the Moon, not around it. ARES is Phase 4: it inherits a mature vacuum-world kit (SELENE, Phase 1; fleet, Phase 3) and proven atmosphere subsystems (Chimborazo, Phase 2).

6. The Build Order

The lightning flashes before the thunder is heard: SELENE launches silently in vacuum before BRONTE shakes the air on Earth.

Phase	System	What gets built	What it proves / earns
1	SELENE PoC (Moon)	1–3 km surface track, 50–100 g, south-pole ridge	2.9 km @ 100 g clears lunar escape; exports water/regolith/metal to cislunar buyers from day one; grows to 10–12 km (30 g) then ~50 km (3 g)
2	Chimborazo (Earth)	14 km first build of the through-mountain line; evacuated tube, plasma window, aero-shell	Mach ~8.9 test capability at 6,160 m on the equator; DoD contracts fund operations; produces the real cost-per-km number
3	Vacuum-world fleet	SELENE-class kits on Phobos, Deimos, Ceres, the Belt	The core bet pays out: Mars' moons become freight terminals before ARES exists
4	ARES (Mars)	~180 km track, featherweight tube	Mars becomes a port; the freight network closes
5	BRONTE full (Earth)	Extend Chimborazo west to ~38 km (Mach 12.6); beyond that, the 1,000 km-class 3 g injector if economics prove out	The endgame — built only on validated cost data

SELENE is ~300× shorter than any full Earth injector and skips every atmosphere subsystem. Chimborazo Phase 1 is ~70× shorter and proves the hard atmosphere parts before any large commitment. They de-risk in parallel.

7. How the Network Moves Mass

Three launchers, one supply chain. Once they exist, moving mass around the inner system mostly costs what running the launchers costs.

- **Earth sends what only Earth can build** — machines, electronics, medicine. The eventual full injector runs 80 t/day (29,000 t/yr); the Chimborazo demonstrator's job is to earn that machine the right to exist. (Rev 1 wrongly pinned the 80 t/day cadence on the demonstrator itself.)
- **The Moon sends raw mass** — propellant feedstock, shielding regolith, structural metal, fired at 2.4 km/s toward cislunar catch points. At full cadence the cislunar propellant supply stops being constrained by what Earth can lift.
- **Mars closes the return trip** — ISRU methalox, samples, and exports ride ARES to Phobos/Deimos depots on electricity.
- **SEP tugs connect the nodes** — weeks per leg, nearly propellant-free; chemical engines do only final descents. The launchers do all the heavy lifting off every surface.
- **Each node funds the next.** Rockets deliver SELENE's first coils (the last rocket buy the lunar build needs); SELENE exports fund Chimborazo; Chimborazo revenue and data fund the fleet; the fleet builds ARES. Every airless body added makes the whole network richer.

8. The Skyhook: How Cost Falls Toward \$5/kg

A rotating tether catches pods and flings them higher. No propellant. More payload per launch.

A long tether rotates in low orbit; its bottom tip sweeps backward, moving slower than orbital velocity. A pod that arrives at the tip gets grabbed; half a rotation later it is released up high, moving faster — to the Moon, GTO, or wherever the mission calls. With a 1.5 km/s tip speed, the launcher needs 6.3 km/s instead of 7.8: ~35% less energy, a slower exit (less heating), and the kick stage shrinks from ~20% of pod mass to 2–5% for trim — the freed 15–18% becomes payload. That shift is where \$5/kg comes from.

- **Keeping it up:** electrodynamic reboost (current through the tether against Earth's field — demonstrated physics, no propellant) plus momentum returned by catching descending capsules.
- **The lunar extension:** a cislunar rotovator catches SELENE pellets and redirects them to LLO, L1, or Earth-transfer; an Earth-orbit tether receives. Cargo crosses cislunar space without burning its own propellant at all.
- **Provenance:** momentum-exchange tethers date to the 1970s; the Boeing/Tethers Unlimited HASTOL study (NASA NIAC, 1999–2001) detailed a 600 km tether with 3.5 km/s tip speed catching Mach-10 aircraft-launched payloads. Materials and manufacturing at scale, not new physics.

9. If the Full Earth Injector Gets Built

Not the plan — the vacuum fleet is the plan. But if the fleet works, this is what becomes possible.

A full Earth injector — ultimately the 1,000 km-class, 3 g, 8 km/s human-rated machine — is the most consequential infrastructure ever proposed. The key point: if SELENE works in vacuum and Chimborazo validates the atmosphere subsystems and the cost-per-kilometer, the full injector stops being a physics question and becomes a construction question.

What \$5–20/kg to orbit changes

- **Satellites:** launch stops dominating cost — a 2,000 kg satellite rides for ~\$20K at \$10/kg vs the \$83–97M a GPS III mission actually paid. Constellations replace on maintenance schedules.
- **Stations:** ISS-class logistics (~\$3.1B/yr, ~\$1.8B of it transport) collapses ~300×; thousand-person stations become economically normal.
- **Orbital manufacturing:** ZBLAN fiber, crystals, alloys — Flawless Photonics drew ~12 km of ZBLAN on ISS in 2024 (1,141 m in one day vs a 25 m prior record); the candid caveat is that published proof of reproducibly superior fiber is still pending. The queue is real either way, and it is waiting on freight prices.
- **Deep space:** with cheap Earth-to-orbit plus lunar propellant, Mars missions stop being nation-sized budget exercises; the Belt becomes commercially reachable.

The strategic picture. The operator of an 80 t/day injector controls the access ramp to space — 29,000 t/yr can replenish a damaged constellation in hours. The host nation stops paying for space access and starts charging for it. The vacuum fleet pays for itself; the Earth injector is what the fleet eventually makes possible.

10. Zero New Physics

Every subsystem has flown, fired, or been demonstrated. The opportunity is assembly, not invention.

Building block	Status — verified June 12, 2026
EMALS (US Navy)	Operational on USS Gerald R. Ford since 2017 incl. combat deployment; 484 MJ flywheel storage, 45 s recharge. CVN-79 (Kennedy) completed builder's trials Feb 2026, delivery ~Mar 2027. 2023 AFOSR-filed report recommends evolving EMALS for lunar launch.
Superconducting maglev	603 km/h crewed record (JR Central L0, 2015) still stands; Shanghai Transrapid commercial since 2003; Chūō Shinkansen now expected NET ~2034 after the Shizuoka settlement (early 2026).
REBCO superconductors	$T_c \approx 92$ K. All-REBCO records: 26.86 T (ASIPP, 2024) → 28.2 T (CHMFL, 2025); all-superconducting record 35.1 T (ASIPP, Sept 2025); 45.5 T hybrid (NHMFL, 2019). CFS demonstrated 20 T SPARC-class coils (2021); SPARC assembly underway (first TF coil placed Jan 2026), first-plasma targeted 2026, $Q > 1$ targeted 2027; ARC (Virginia) early 2030s. $J_c > 1,000$ A/mm ² at 30 T at 4.2 K. Launch LSM needs only 5–10 T at the gap.
MHD plasma window	Invented by A. Hershcovitch (Brookhaven), patented 1995; holds ~9 atm; demonstrated for non-vacuum e-beam welding at cm-scale apertures. Scaling to the Ø4 m bore (2.8× baseline power) is the program's R&D centerpiece — Phase 1 tests it at full bore.
X-59 shock shaping	First flight Oct 28, 2025; first supersonic flight June 5, 2026 (Mach 1.1); Mach 1.4 acoustic points next; XVS camera-vision cockpit flight-proven.
High-g electronics	Gun-launched guidance routinely qualified to 15,000–20,000 g (Excalibur-class). SELENE's 100 g cargo mode is conservative by two orders of magnitude.
EM launch-assist market	Auriga Space: \$12.2M cumulative incl. AFWERX D2P2; hypersonic-test-as-revenue model proven in market; outdoor track (Thor) planned 2026.
Orbital manufacturing demand	Flawless Photonics: ~11.9 km ZBLAN drawn on ISS (Feb–Mar 2024; 1,141 m/day vs 25 m prior record); Luxembourg production expansion; published performance characterization still pending — flagged honestly.
Lunar mass-driver physics	O'Neill, NASA Ames studies 1976–77; Mass Driver One demo 1977. SJSU sizing study (2023): 25.4 kg pellets, 2.4 km/s, 10 s cadence, 8.7 MW polar solar, ~1.0 MA per bucket coil.
Lunar intent at scale	SpaceX lunar pivot + mass-driver statements (Feb–Mar 2026). Founder remarks, not a program of record — but the direction of the largest launch company on Earth.
Fusion pull (He-3 context)	NIF ignition Dec 2022 (3.15 MJ); best shot now 8.6 MJ, target gain >4 (Apr 2025), ignition repeated 9+ times. Commercial He-3 demand is already contracted: Interlune–Bluefors offtake (Sept 2025) for up to 10,000 L/yr from 2028, prospecting mission 2027.

11. The Hard Parts

A document about a project this large that doesn't name the real problems is salesmanship, not engineering.

- **Plasma window scaling.** Centimeter demos → Ø4.0 m bore at 2.8× baseline power. First R&D dollar; if it doesn't scale, the muzzle needs a different answer and we must learn that at demonstrator scale, cheaply.
- **Cost per kilometer is unknown.** The \$20–250/kg projection lives or dies on vacuum-rated superconducting track cost. Nobody has published a credible figure. Chimborazo Phase 1 produces the first real one.
- **The El Arenal piers.** 110–590 m A-frames in a 0.4 g seismic zone — beyond Millau's 343 m record. Partial trenching, rock-buttressed spans, or accepting a lower exit are the studied outs.
- **The reserve.** Chimborazo sits inside a protected fauna reserve under a rights-of-nature constitution. The legal path may be harder than the rock. Stated plainly because pretending otherwise would be fatal later.
- **SELENE's first bill is a rocket bill.** First coils, power, and radiators ride to the Moon at today's prices; the starter track must fit one or two heavy-lift flights.
- **Catching is harder than throwing.** Until the first cislunar catch happens, export revenue is theoretical. Wide-cone tugs first, tethers later.
- **Polar ice uncertainty.** Shackleton-specific reserves are unproven (bounded $\leq 5\text{--}10$ wt% in surveyed regolith). The early business plan must close on regolith, oxygen, and metal even in the lean-ice case. Chang'e-7 data lands within a year.
- **The treaty question.** The Outer Space Treaty is silent on commercial mass drivers; a kinetic launcher is inherently dual-use. That ambiguity buys scrutiny on one side and AFWERX funding on the other. EMALS and GPS both walked this road.

12. What KERAUNOS Unlocks

Orbital manufacturing

- **ZBLAN fiber** — up to 100× lower theoretical loss than silica; ~12 km drawn on ISS in 2024; customers queue on freight prices. KERAUNOS is the freight price. (Performance proof still pending publication — the bet is stated, not oversold.)
- **Orbital bioprinting** — organs that collapse under their own weight in 1 g; daily return pods turn it from stunt to logistics business.
- **Debris salvage** — thousands of tonnes of aerospace alloy already in orbit becomes ore when tug-hours are cheap.

Geopolitical position

- **The Space Panama Canal** — the host nation stops paying for access and starts charging for it.
- **Presence as a supply chain** — 80 t/day secures orbital slots and lunar territory as a logistics fact, not a flag.
- **The deterrence dividend** — constellation replenishment in hours, not months.

Fusion economy

- **The He-3 supply chain.** Lunar regolith holds an estimated (model-based) ~1M tonnes of He-3. D-He3 fusion is **low-neutron** (D-D side reactions keep it from being strictly aneutronic) — far less activation, direct conversion potential. At today's ~\$20M/kg (~\$20B/tonne) market price and SELENE's ~\$5/kg lunar launch cost, delivery economics are absurdly favorable; Interlune's contracted offtakes (Bluefors, 2028–2037) prove demand precedes the fusion era. SELENE isn't just a launcher — it's the fuel logistics for the energy industry after next.
- **The energy loop.** Trackside storage doubles as grid-scale battery between launch windows; the same plant powers debris-sweeping lasers over the corridor. Nothing idles.

Verification note: all external claims in this revision were re-checked against primary reporting and literature on June 12, 2026; all physics re-derived from first principles the same day. Annex A (BRONTE Chimborazo Design Map) is bound after Section 4. The SELENE Design Map (KER-SEL-TRK-002) is issued as a companion sheet.